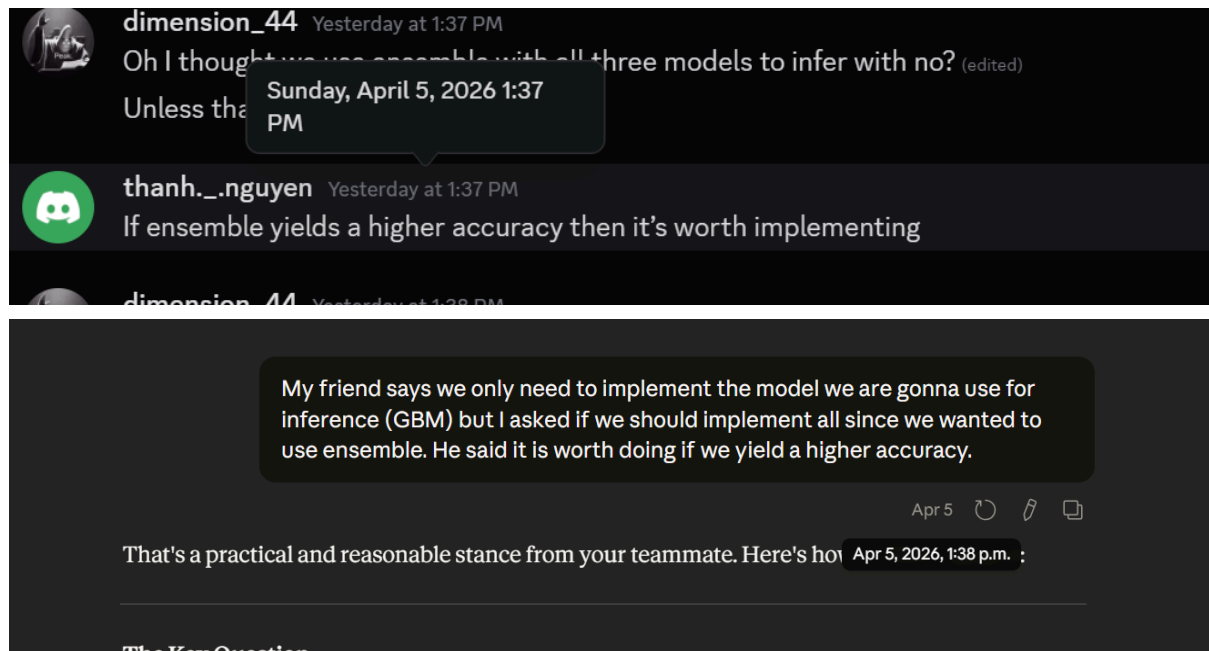


CSC311 ML Challenge - AI usage

Note on AI usage from Yash Khanchandani

Here is a link to my chat with Claude: [Painting classification with gradient boosting - Claude](#)

Note: We used Claude mainly to consult our decisions on each model, and also used it to further understand and verify our findings on the performance issues faced by the models on the predictions.



In the first image, Thanh suggested implementing Naive Bayes and Logistic Regression iff the ensemble accuracy is higher than GBM. This was mentioned on Sunday April 5 at 1:37 PM. In Claude's response it suggested a script to test out the accuracies of the individual models and the majority-vote ensembles as well.

Note 2: To meet the requirement of exploring at least three model families, our team decided to implement Naive Bayes alongside Gradient Boosting and Logistic Regression. While the LLM provided an initial implementation for GaussianNB, **the decision to include it was based on our pre-processing step where we normalized continuous features.** I independently verified that the Gaussian assumption was a reasonable approximation for our **standardized numerical data** before including these results in the final report.

Final Note: I used Claude as a discussion board to reason out possibilities for model performance results. While Claude did provide an initial draft of the possible trade-offs, I found the reasoning to be too generic, so I discarded the draft and wrote the naive bayes section of the report myself, specifically highlighting the 70/77 categorical features vs the 7/77 numerical features, as well as the information loss caused by the binarization of the numerical features caused by the Bernoulli NB.